



Overclock your graphics card

If you're giving up on your graphics card, you might as well try and overclock it!

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So you got yourself a spanking new graphics card and were excited about playing the latest games at the highest resolutions—only to find it faltered while running a low-end configuration of *Doom 3*? If that made you give up on your card, you might as well risk overclocking it to see if you can get that last bit of speed out of it.

What You Need

In this workshop, we tell you how to go about overclocking your nVidia card using a simple utility called nVHardPage and enabling some hidden menus in the default driver's menu for the card. You can download nVHardPage from www.guru3d.com/nvhardpage/

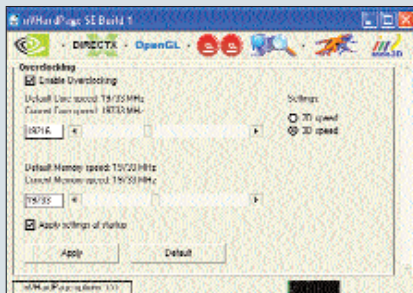
1 Running The Software

The software does not require any installation. Simply navigate to the folder where you unzipped the files, and find a file called 'nVHardPageSE'. Running the file will present you with a tweaking menu.

The default screen shows the system configuration including those of the graphics card, the CPU and the OS.

2 The Software Interface

At the top, you'll notice icons for nVidia (card), DirectX and OpenGL, and an icon comprising two speedometers, which stands for the inbuilt overclocking utility. This utility has two sliders, which reflect the core clock speeds of the card and the accompanying video RAM speeds. They need to be adjusted together—your card should not run faster than the video RAM. Of course, both these speeds have thresholds you need to adhere to.



Move the slider to get optimum performance out of the card



The Control Panel

We won't be messing around with the card from within this software, but will do it from within the nVidia drivers, as they give us certain options that the software doesn't—namely, the automatic overclocker! Note that you can use nVHardPage to overclock your card if you want to, but we don't recommend that you do. nVHardPage gives you too much leeway—you could easily damage your card by moving the slider too far to the right! Why we used nVHardPage at all is because running it makes the overclocking options in the nVidia drivers available.

Now, to un-hide the frequency controller slider for the driver interface, click on the nVidia icon at the top and then click on the control panel set-



Enable various settings using the control panel

tings. Here you'll find a list of options that signify the settings you can tweak from within the nVidia driver interface. The unchecked boxes signify the hidden tweaks. Now here, tick the 'Enable Coolbits' box. This will ensure that you get a speed slider when you open the display properties next time.

There are loads of other options here, such as AGP settings, which lets you choose the AGP acceleration, and a troubleshooting page—along with a page for configuring Direct 3D rendering, where you choose how many frames the card renders in advance. This can be tricky, because if you play a FPS you never know what kind of frames are required next—so if you render them in advance, it could prove to be a bottleneck.



The Overclocking Menu

Now since we've enabled the overclocking feature in the drivers, let's try and overclock the card!

To go to the nVidia settings page,

right click on the desktop. Choose Properties > Settings > Advanced, and then click the tab that has your graphics card's name.

A menu opens up on the left with a multitude of options and settings. Click on the option marked 'Clock Frequency Settings'. On the right, select the radio button labelled 'Manual Overclocking'. Selecting it will present you with a disclaimer. Read it and hit the 'I accept' button. You will be presented with two sliders, one for 'Core Clock Frequency' and the other for 'Memory Clock Frequency'.

5 Overclocking The Card

You could, of course, go ahead and overclock the card by simply sliding the slider as far to the right as you think is safe, but if you take the slider too far to the right, you could render your card forever useless.

There is an easy way to test the potential limit of your card. In the drop-down menu saying '2D Performance', select '3D'. A button labelled 'Detect optimal Frequencies' gets enabled. Clicking it will instruct the drivers to take all system parameters into consideration and give you the optimum frequencies for both the GPU core and the memory frequency.

There is, however, no guarantee that the card will run all games at these settings. Once the card gives you these settings, you should test the system stability by running different games.

If all goes well, try increasing the frequency a little, and test again. Continue this process until you get the maximum value at which your system stays stable. This is the optimum overclocked speed of your card, and this can be above or below the auto-detected settings.

If you have an ATi card, you can download some utilities from www.voodoofiles.com/9742

These software work similar to the one we've talked about, and you shouldn't have too much trouble overclocking your ATi card. ■



Set the core and memory clock frequencies here